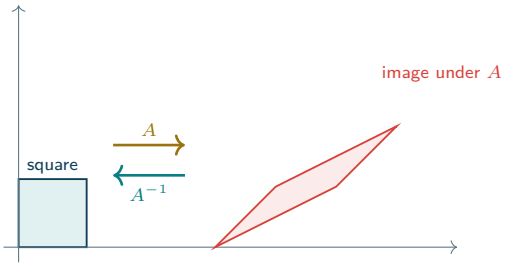
LINEAR ALGEBRA LADDER 4 OF 9

Inverses and elementary matrices

An inverse undoes a matrix, and the row operations of elimination are themselves matrices.

1 / THE RULE AND THE PICTURE

A^{-1} is the matrix that undoes A, so $A^{-1}A = AA^{-1} = I$



For $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ with $\det A = 2 - 1 = 1$ the inverse swaps the diagonal, negates the other entries and divides by the determinant:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \tfrac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

so $A^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$, and an invertible map is reversed by its inverse.

2 / SOLVING WITH THE INVERSE

$Ax = b$ has the solution $x = A^{-1}b$ whenever A^{-1} exists

For $b = (5, 3)$ we get $x = A^{-1}b = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 5 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$, and indeed $2 \cdot 2 + 1 = 5$ and $2 + 1 = 3$.

The same inverse solves every right-hand side at once, which is why it is worth computing when many b occur, although elimination is cheaper for a single system.

3 / ELEMENTARY MATRICES / ELIMINATION AS MULTIPLICATION

Each row operation is multiplication by an elementary matrix, and the product of them is A^{-1}

Reducing $[A \,|\, I]$ to $[I \,|\, A^{-1}]$ takes two steps:

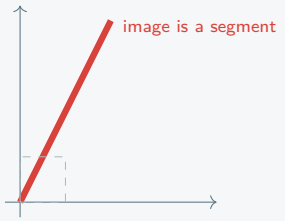
$$R_1 \leftarrow R_1 - R_2 : \; E = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}, \; EA = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

$$R_2 \leftarrow R_2 - R_1 : \; F = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}, \; FEA = I$$

Hence $A^{-1} = FE = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$, found by the same row operations applied to the identity beside A.

4 / WHEN THERE IS NO INVERSE / SINGULAR MATRICES

A matrix with $\det = 0$ squashes the plane and cannot be undone



For $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ the determinant is $4 - 4 = 0$ and the whole plane lands on the line through $(1, 2)$, so many points share each image and the map cannot be reversed.
Elimination shows the same thing as a zero row, and $Ax = b$ then has no solution or infinitely many.
For products the order reverses, $(AB)^{-1} = B^{-1}A^{-1}$, as in taking off shoes before socks.

ASIDE / INVERSE OF A PRODUCT

Undoing two steps means undoing the last one first, which is why $(AB)^{-1} = B^{-1}A^{-1}$.

LINKS TO OTHER SHEETS

Elimination is in *The LCM method as Gaussian elimination*; the previous sheet is *Matrices as transformations* and the next is *Determinants: area, volume and invertibility*.

Convention. I is the identity matrix; E, F are elementary matrices; $[A \,|\, I]$ is A augmented by the identity.